ARCHITECTURE COURSE · GQM WORKSHOP

GQM across the course books

How five architecture books' concepts map onto Goal–Question–Metric (GQM). Find the book you read, keep its vocabulary, and use the links to GQM in your group.

GQM: Basili & Weiss (1984); Basili, Caldiera & Rombach (1994). Define top-down, interpret bottom-up.

Mapping table

GQM	Building Evolutionary Architectures	The Art of Systems Architecting	Continuous Architecture in Practice	Computer Networks	Software Architecture in Practice
Goal	Architectural characteristic	Client purpose, value judgement	Quality attribute focus	QoS target, at most	Business goal, quality attribute
Question	Holistic fitness function	Heuristics; purpose turned into criteria	Which decision is pending?	Where is delay or loss caused?	Scenario stimulus and environment; tactics
Metric	Atomic fitness function	Performance model, certification criteria	Operational and feedback-loop data	Throughput, RTT, jitter, loss	Response measure
Select	Which checks to automate	Satisficing	Delay until data justifies it	Exact metric definitions	Utility tree ranking
Feedback	Emergent fitness functions, pipeline	Progressive refinement	Feedback loops	Network monitoring	Evaluation, ATAM risks

Per book

Building Evolutionary Architectures

Ford, Parsons, Kua & Sadalage

Concepts: architectural characteristics, fitness functions (atomic/holistic, triggered/continuous, static/dynamic, automated/manual, intentional/emergent), incremental and guided change, architectural quantum, connascence, deployment pipelines, automated governance.

- A fitness function is a metric with a threshold, run automatically. GQM supplies the goal that says why it exists.
- Atomic fitness functions are single metrics; holistic ones answer questions that combine several.
- Automating a check moves a metric from "collect" to "have".
- Emergent fitness functions are GQM's feedback step.

The Art of Systems Architecting

Maier & Rechtin

Concepts: client, purpose orientation, value judgements, ill-structured problems, heuristics, views and models, progressive refinement, certification, satisficing.

- Purpose orientation is the goal; the client is the goal's viewpoint.
- Turning a qualitative purpose into testable criteria is the Goal → Question → Metric chain.
- Certification needs agreed measures; fuzzy goals make subjective metrics legitimate. Satisficing is the selection step.

Continuous Architecture in Practice Erder, Pureur & Woods

Concepts: six principles (quality attributes first, delay decisions, architect for change, architect for build/test/deploy/operate, …), decision logs, QA scenarios and utility trees, technical debt, feedback loops, fitness functions.

- “Focus on quality attributes” is the workshop’s starting point: one NFR as the goal.
- Delaying a decision works only if you know which data would settle it; the cost column says if waiting is affordable.
- Feedback loops are GQM’s bottom-up half. Log which question each decision depends on.

Computer Networks Tanenbaum, Feamster & Wetherall

Concepts: bandwidth vs throughput, delay components, RTT, jitter, packet loss, bandwidth-delay product, QoS, congestion and flow control, protocol layers, network management (SNMP).

- Supplies the metric level: exact definitions, units, tools.
- QoS parameters are ready-made metrics; delay breakdown and layers structure “where is it caused?” questions.
- Network metrics are often proxies for the user-facing goal.

Software Architecture in Practice Bass, Clements & Kazman · closest fit

Concepts: quality attributes, QA scenarios (source, stimulus, artifact, environment, response, response measure), tactics, utility tree, ATAM (sensitivity and trade-off points, risks), QAW, business goals.

- Business goal + quality attribute = the GQM goal; the scenario’s environment is the goal’s context.
- The response measure is a ready-made metric; a good scenario is nearly a question–metric pair.
- Utility tree importance ≈ question value (its difficulty is about building, not the cost of data). Tactics give cause questions.

Extra workshop task (optional)

Translate one picked question–metric pair into your book’s terms: a QA scenario (SAiP / Continuous Architecture), a fitness function (Evolutionary Architectures), a precise metric definition (Computer Networks), or a certification criterion (Art of Systems Architecting).

Example: one metric, three ways

- **GQM:** “How fast is checkout at peak?” → p95 latency per endpoint.
- **QA scenario:** 500 users/s place orders during peak; the order API responds; p95 under 800 ms.
- **Fitness function:** nightly load test fails the build if p95 exceeds 800 ms.